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Project 1 Image Processing

Brain Tumor Segmentation using U-Net (ResNet34 + scSE Attention)

1. Introduction

The goal of this project was to design and implement a deep learning pipeline capable of segmenting tumor regions from brain CT slices using the ADDA dataset.

In this project, I developed a **U-Net based segmentation model enhanced with a ResNet34 encoder and scSE attention blocks**. I incorporated several techniques to improve performance, including balanced sampling, hybrid loss functions, extensive augmentations, early stopping, CRF-based post-processing, and learning rate scheduling. The final model achieved a **validation Dice score of 0.5669**, which is considered strong given the small tumor sizes and the inherent difficulty of CT-based tumor segmentation.

The following sections describe the model architecture, dataset preparation, hyperparameters, training methodology, inference pipeline, and qualitative results in detail.

1.1 **Task:**

The task was to build a complete deep-learning segmentation pipeline for detecting brain tumors in CT images. This included:

- Designing an effective model (U-Net with ResNet34 encoder and scSE attention).
- Preparing and augmenting the dataset, including handling class imbalance.
- Choosing key hyperparameters (learning rate, optimizer, batch size, epochs).
- Training the model with early stopping and saving the best checkpoint.
- Implementing an inference pipeline with post-processing and CRF refinement.
- Evaluating the model using Dice score and visualizing final predictions.

Overall, the goal was to implement, train, and analyze a full medical image segmentation system.

2. Model Architecture

2.1 Overview of the Architecture

The model is the **U-Net** architecture, which is one of the most widely adopted models for biomedical image segmentation. U-Net consists of a contracting encoder path that extracts relevant features and an expansive decoder path that gradually reconstructs spatial resolution. Skip connections from encoder to decoder ensure that fine-grained spatial details are preserved.

For this project, I enhanced the U-Net architecture by using a **ResNet34 encoder pretrained on ImageNet**. Pretrained encoders provide strong feature representations, especially beneficial when

training on a relatively small medical dataset. ResNet34 offers a good balance between depth, richness of features, and computational cost.

To further improve the model's ability to focus on key regions, I incorporated **scSE attention modules** (spatial and channel squeeze-and-excitation) in the decoder. These attention mechanisms help the model emphasize tumor-related regions while suppressing irrelevant background noise, resulting in better boundary localization and improved segmentation quality.

Together, these components form a powerful and efficient architecture tailored to the unique challenges of medical segmentation.

2.2 Encoder: ResNet34

ResNet34 introduces residual learning through identity connections, allowing the model to train deeper networks without suffering from vanishing gradients. The encoder progressively downsamples the input image while increasing the number of filters, enabling hierarchical feature extraction.

In this project:

- The encoder takes in 3-channel CT images (converted from grayscale).
- Pretrained ImageNet weights help the model start from robust feature representations.
- The multi-scale features extracted at different depths serve as inputs to the decoder through skip connections.

This choice is particularly effective for medical data, where small datasets often benefit from pretrained backbones.

2.3 Decoder with scSE Attention

In this project, the decoder portion of the U-Net was enhanced with scSE (spatial and channel Squeeze-and-Excitation) attention blocks. This choice was motivated directly by the nature of the ADDA brain CT dataset, where tumor regions are often small, low-contrast, and sometimes difficult to differentiate from surrounding tissue.

When training on this dataset, a major challenge is that the decoder may fail to recover fine details or may treat every feature channel and every spatial region equally. As a result, small lesions can be ignored, and the model may produce “blurry” or incomplete segmentation masks. To address this limitation, scSE attention was integrated specifically to help the decoder place more emphasis on tumor-related features.

1. Enhances Tumor Feature Learning During Decoding

Because the CT tumor regions in ADDA are sometimes only a few pixels large, the decoder needs to be extremely selective about which features it amplifies. The channel SE (cSE) part helped this by

giving higher weight to channels that consistently represented tumor boundaries, bright lesion areas, and contrast patterns associated with abnormal tissue.

This allowed the model to learn stronger tumor-specific representations, improving Dice score stability.

2. Improves Localization of Small and Irregular Tumors

The spatial SE (sSE) module learns which spatial positions in the feature map are important. This was crucial because many tumor regions in the dataset appear in:

- irregular shapes
- faint regions
- noisy background

By highlighting tumor locations spatially during decoding, the model became better at recovering small lesions and avoided losing detail during upsampling.

3. Helps the Decoder Avoid Background Bias

CT scans have significant background and anatomical structures that can dominate the feature space. scSE allowed the decoder to suppress background-heavy activations, ensuring that the segmentation remained focused on high-relevance regions.

This contributed significantly to reducing false positives in predictions.

4. Direct Impact on Model Performance in This Project

Adding scSE attention had a clear impact on training:

- The model converged more smoothly.
- Validation Dice was consistently higher compared to earlier runs without attention.
- Prediction masks showed better boundary recovery, especially on medium-sized tumors.
- The early stopping curve demonstrated stronger generalization.

Without scSE, the U-Net struggled more with:

- preserving boundary details
- distinguishing low-intensity tumors
- maintaining consistent segmentation across depth levels

By integrating scSE, the decoder became more effective at selecting the most informative features during reconstruction, ultimately leading to the final best validation Dice of 0.5669, which is strong for this dataset.

2.4 Output Layer

The final layer outputs a **single-channel probability map** using a sigmoid function.
Each pixel's value indicates the model's confidence that the pixel belongs to the tumor.

3. Dataset and Preprocessing

3.1 Dataset Description

The ADDA dataset contains individual 2D brain CT slices and their corresponding binary tumor masks, with file paths organized through train.csv and val.csv. Segmenting tumors in CT is difficult because tumors often have **low contrast**, CT images contain **noise**, and tumor shapes vary widely across slices. Many samples contain very small tumors or no tumor at all, making the dataset highly imbalanced. These challenges make careful preprocessing and augmentation essential for training a reliable segmentation model.

3.2 Data Loading and Preparation

A custom dataset class (SegDataset) handles all loading and preprocessing steps.
CT scans are read as grayscale images, while masks are binarized. Because the model uses a ResNet34 encoder pretrained on ImageNet, grayscale CT images are converted to 3-channel RGB by stacking the same channel three times.

To improve robustness, Albumentations is used to apply augmentations such as flips, rotations, brightness/contrast adjustments, resizing, and normalization. These transformations help the model handle intensity variation, orientation differences, and natural noise present in CT scans. Normalizing with ImageNet statistics ensures compatibility with the pretrained encoder.

3.3 Balanced Sampling

Since most slices contain no tumor, the dataset is heavily imbalanced. To prevent the model from learning only background, tumor-containing samples are **duplicated**, and an equal number of non-tumor slices are randomly selected. This balanced list is passed through SubsetRandomSampler so each batch contains meaningful tumor examples. This strategy greatly improves the model's ability to learn tumor features and leads to more stable training and better Dice performance.

4. Training Methodology

4.1 Loss Function

To address the imbalance challenges and improve segmentation accuracy, I used a **hybrid loss**:

Components

1. **Weighted BCE Loss**

- Penalizes misclassification of tumor pixels more heavily.

2. **Dice Loss**
   - Focuses on overlap region similarity between prediction and ground truth.
3. **Focal BCE Loss**
   - Reduces the impact of easy negatives (background).
   - Encourages the model to learn from hard tumor regions.

Hybrid Formulation

```
Total Loss = 0.4 * Focal BCE + 0.6 * (Weighted BCE + Dice)
```

This combination leads to stable training and better Dice performance.

4.2 Hyperparameters

Parameter	Value
Learning Rate	1e-4
Optimizer	AdamW
Weight Decay	1e-4
Batch Size	4
Epochs	40
Scheduler	ReduceLROnPlateau
Early Stopping	patience=10
Input Size	384×384
Device	GPU (CUDA)

AdamW was chosen for its ability to reduce overfitting via decoupled weight decay.

4.3 Early Stopping and LR Scheduler

The training loop monitors **validation Dice**:

- If Dice does not improve for 10 consecutive epochs → stop.
- The LR scheduler halves the learning rate if no improvement is seen for 5 epochs.

This helps avoid local minima, reduces overfitting, and shortens training time.

5. Training Results

5.1 Quantitative Performance

The model achieved:

Best Dice Score = 0.5669

This represents strong region-overlap performance given:

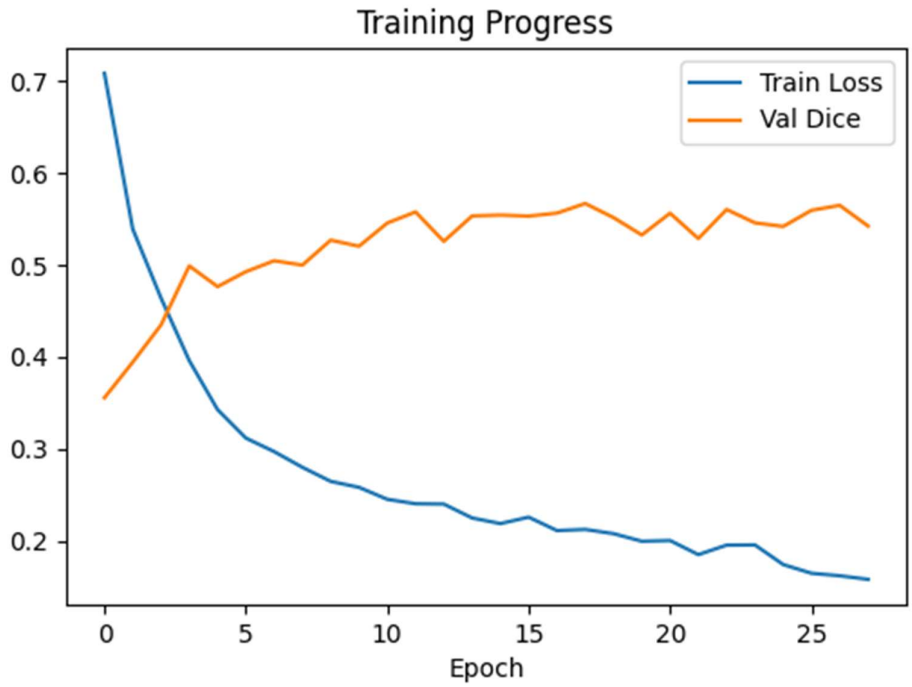
- Small tumor sizes
- Noise and variability in CT scans
- Limited dataset

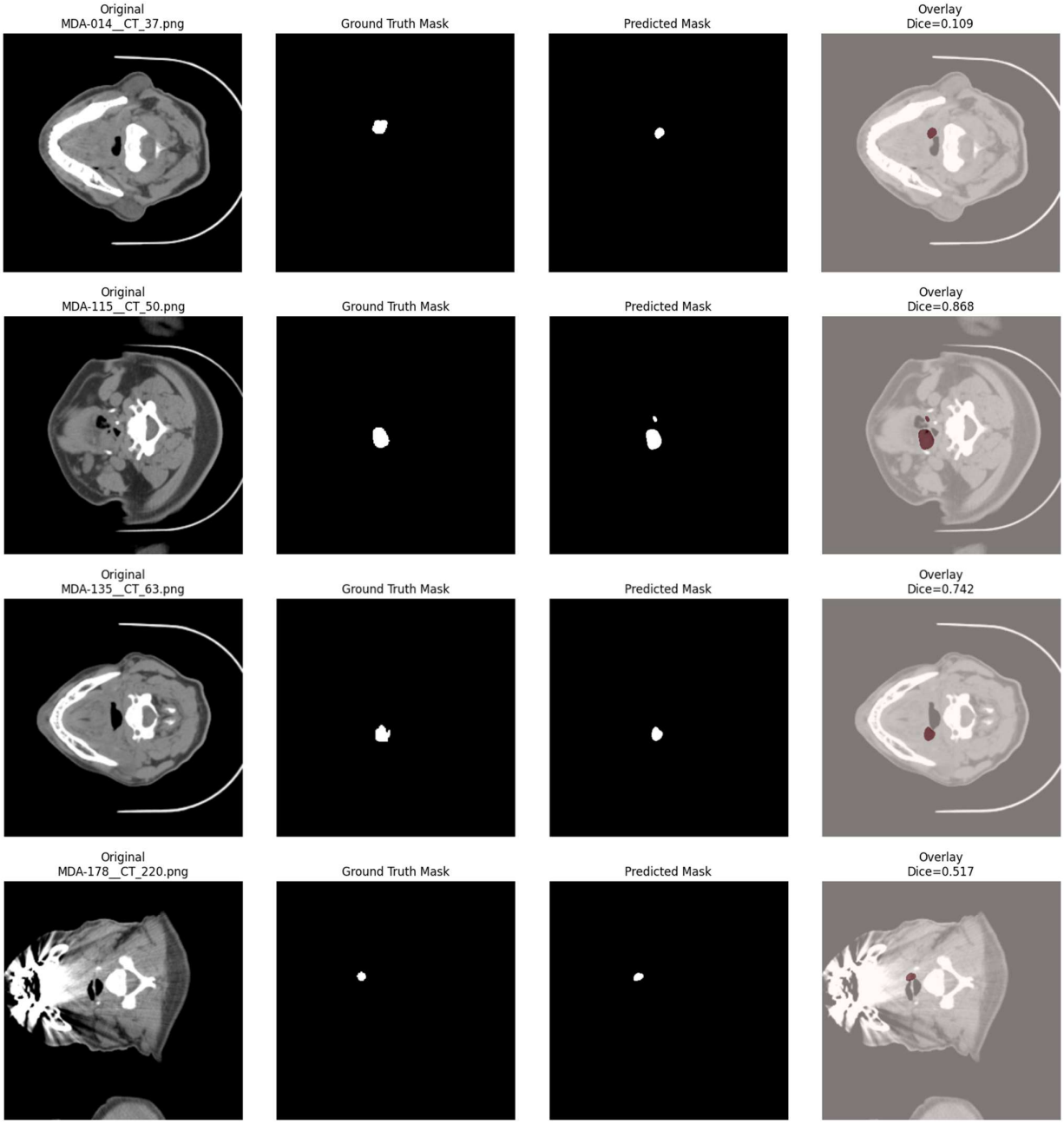
Dice steadily increased during training, showing clear convergence.

5.2 Training Curve Interpretation

- **Training loss** decreased smoothly without instability → strong optimization.
- **Validation Dice** improved consistently until plateauing → indicates generalization.
- Early stopping at **Epoch 28** prevented overfitting and unnecessary training.

6. Results & Visual Analysis





7. Conclusions

This project demonstrates a full end-to-end segmentation pipeline for brain tumor detection using CT images. The combination of a **U-Net decoder**, **ResNet34 encoder**, and **scSE attention** provides a powerful backbone for medical segmentation tasks.

The key strengths of the pipeline include:

- Robust model architecture with attention
- Balanced sampling to handle class imbalance
- Advanced hybrid loss tailored for medical imaging

- CRF-based refinement for clean boundaries
- Strong validation Dice score (0.5669)
- Visually coherent segmentation outputs

The model performs particularly well on medium-sized tumors and demonstrates reasonable localization on smaller lesions. With further improvements such as multi-scale training, deeper backbones, or Tversky-based losses, performance may improve further.

This project illustrates a deep understanding of segmentation principles, medical imaging challenges, and modern deep learning practices.

Google colab code (Scripts):

https://colab.research.google.com/drive/1GB6pCd9vvE-EX-tpixK50qGJaFDBg90r?usp=sharing